Starlet Project Introduction

Tarlan Bahadori
April 13, 2026

UC RIVERSIDE

Workshop Overview

- Introduction to GIS
- → Introduction to Starlet Project
- Hands-On Example
- Introduction to LLM-Assisted Styling
- Competition

UC RIVERSIDE

Motivation

- Geospatial datasets can be **large** and **slow to render**
- Raw data cannot support **interactive maps**
- Need **scalable preprocessing** and **fast visualization**

Install Starlet:

- **Requires Python 3.10 +**

Command to use in terminal:

- **pip install starlet**

More Information:

- **Visit Project GitHub:** https://github.com/tarlaun/starlet
- **Visit Package on PYPI:** https://pypi.org/project/starlet/

UC RIVERSIDE

What Starlet Does:

1. Input **vector** dataset (GeoJSON / GeoParquet)
2. Initial **Partitioning** for Parallel Processing
3. Build **Histograms**
4. Build **Tile Pyramid**
5. Generate **MVT** tiles
6. Serve **Interactive Map**

UC RIVERSIDE

Why use MVT?

Efficient & scalable

- Lightweight, supports large datasets
- Smooth zoom without re-rendering tiles

Smooth interactions

- Fast zoom, pan, rotate, animate
- Rendered dynamically in browser

Dynamic styling

- Style maps at runtime
- No need to regenerate tiles

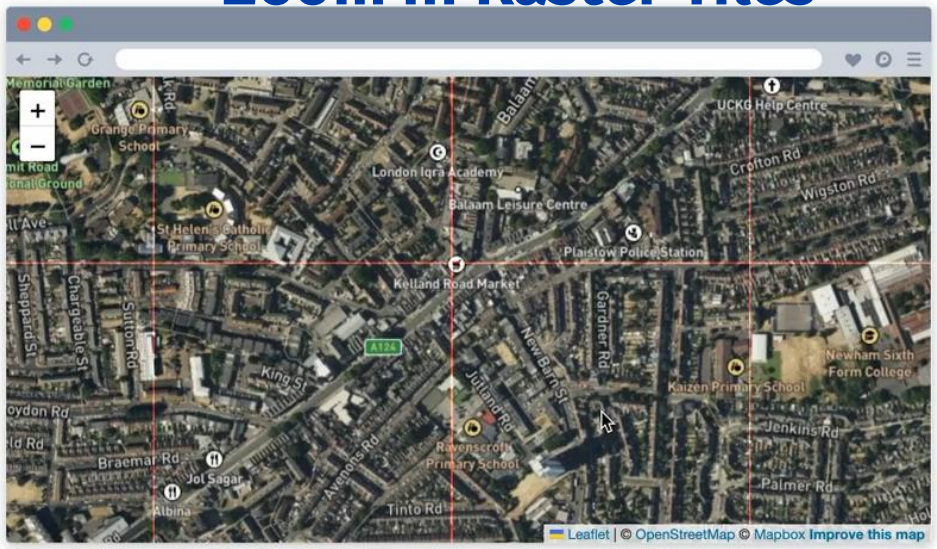
Small size, fast loading

- Reduced bandwidth
- Efficient caching

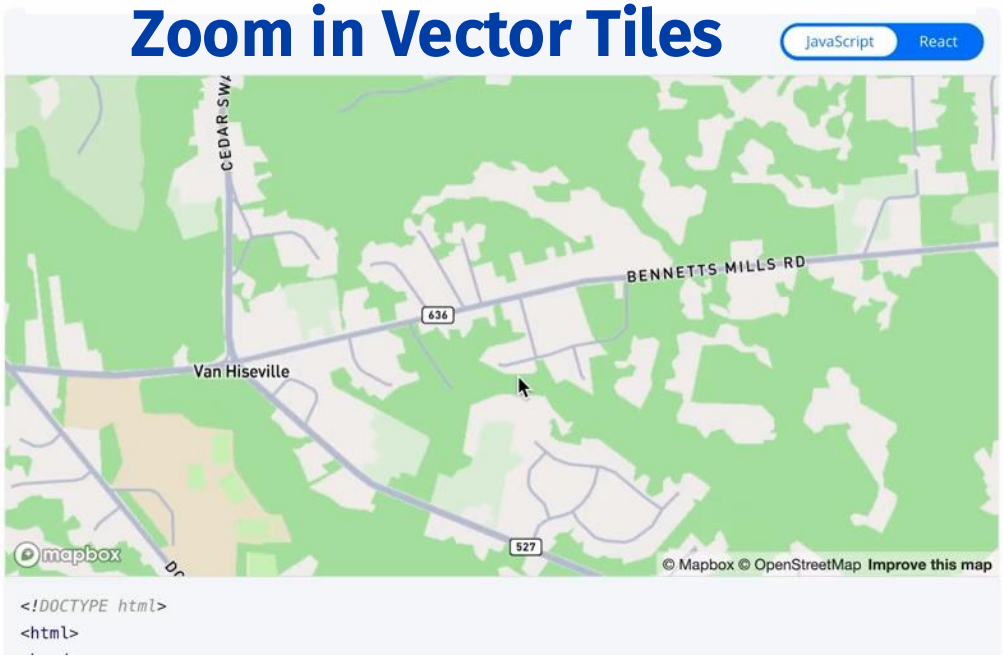
Queryable data

- Access attributes directly
- Enables interactivity & exploration

Zoom in Raster Tiles

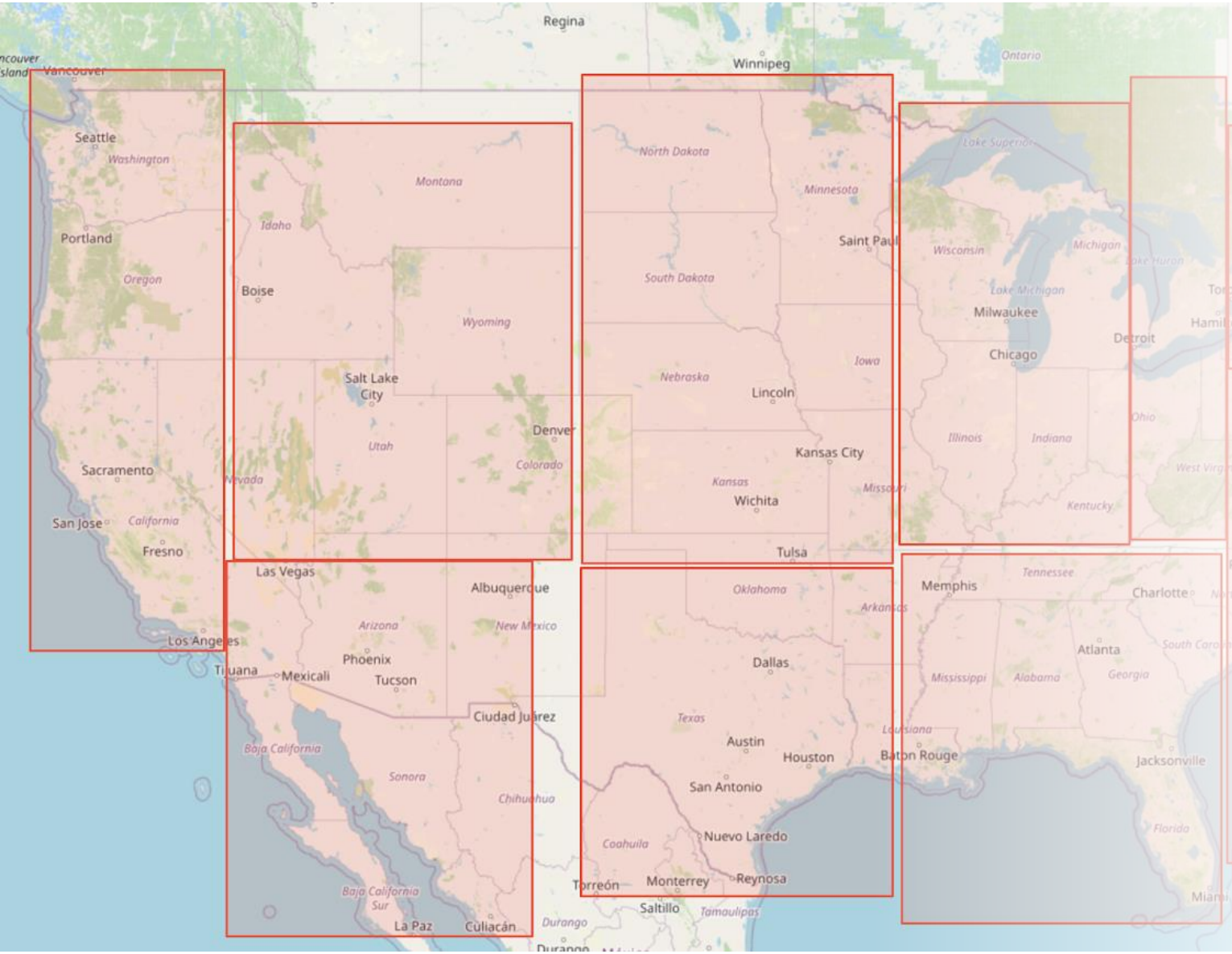


Zoom in Vector Tiles



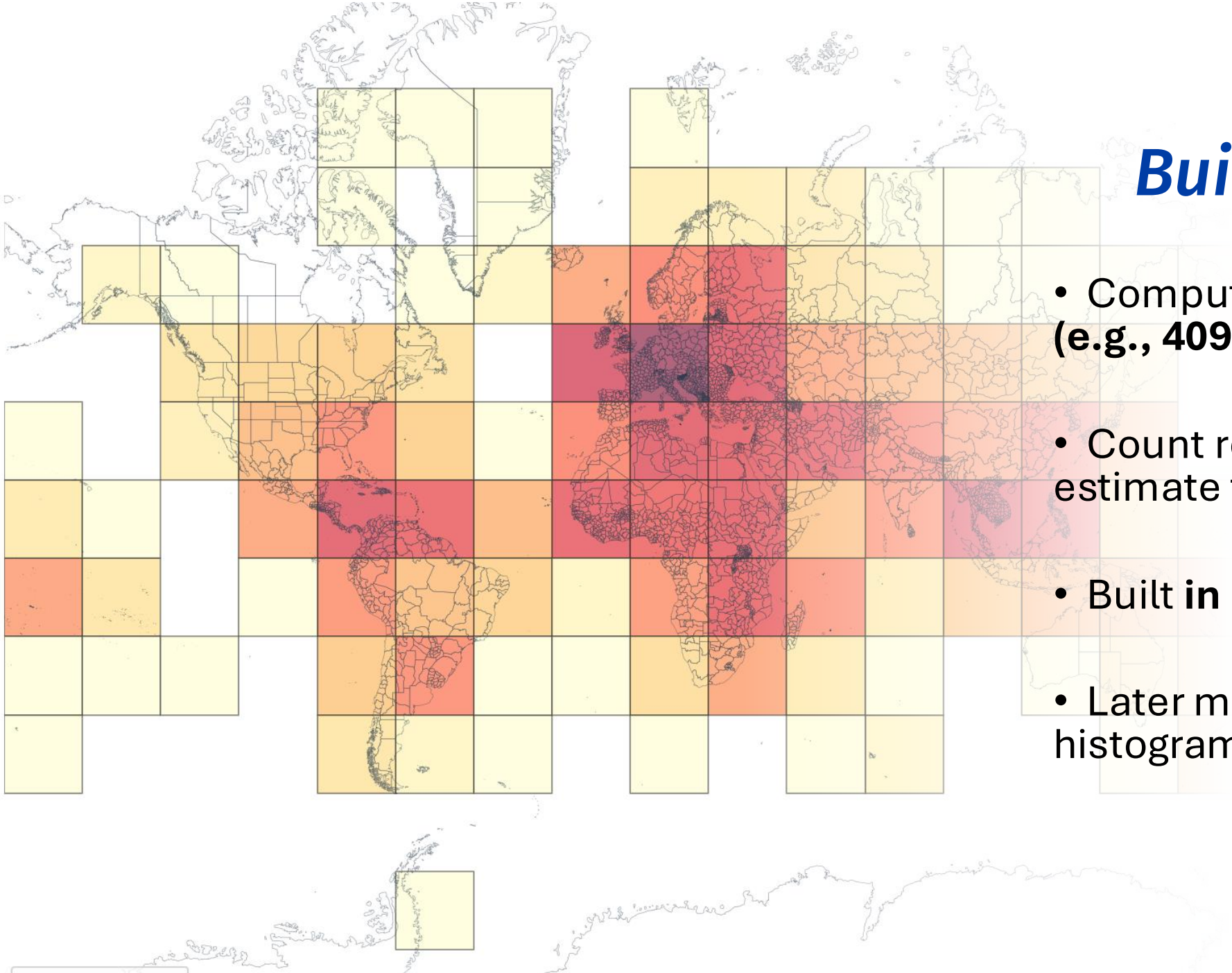
UC RIVERSIDE

Parallel Partitioning of Input Data:



- Divide the input into spatial partitions before tile generation
- Process partitions independently with multiple workers
- Limit the number of active writers to control memory and file handles
- Send excess records to an overflow file and process them in later rounds

UC RIVERSIDE



Build Histograms:

- Compute **spatial density grid (e.g., 4096×4096)**
- Count records per cell (helps estimate tile sizes)
- Built **in parallel per partition**
- Later merged into global histogram

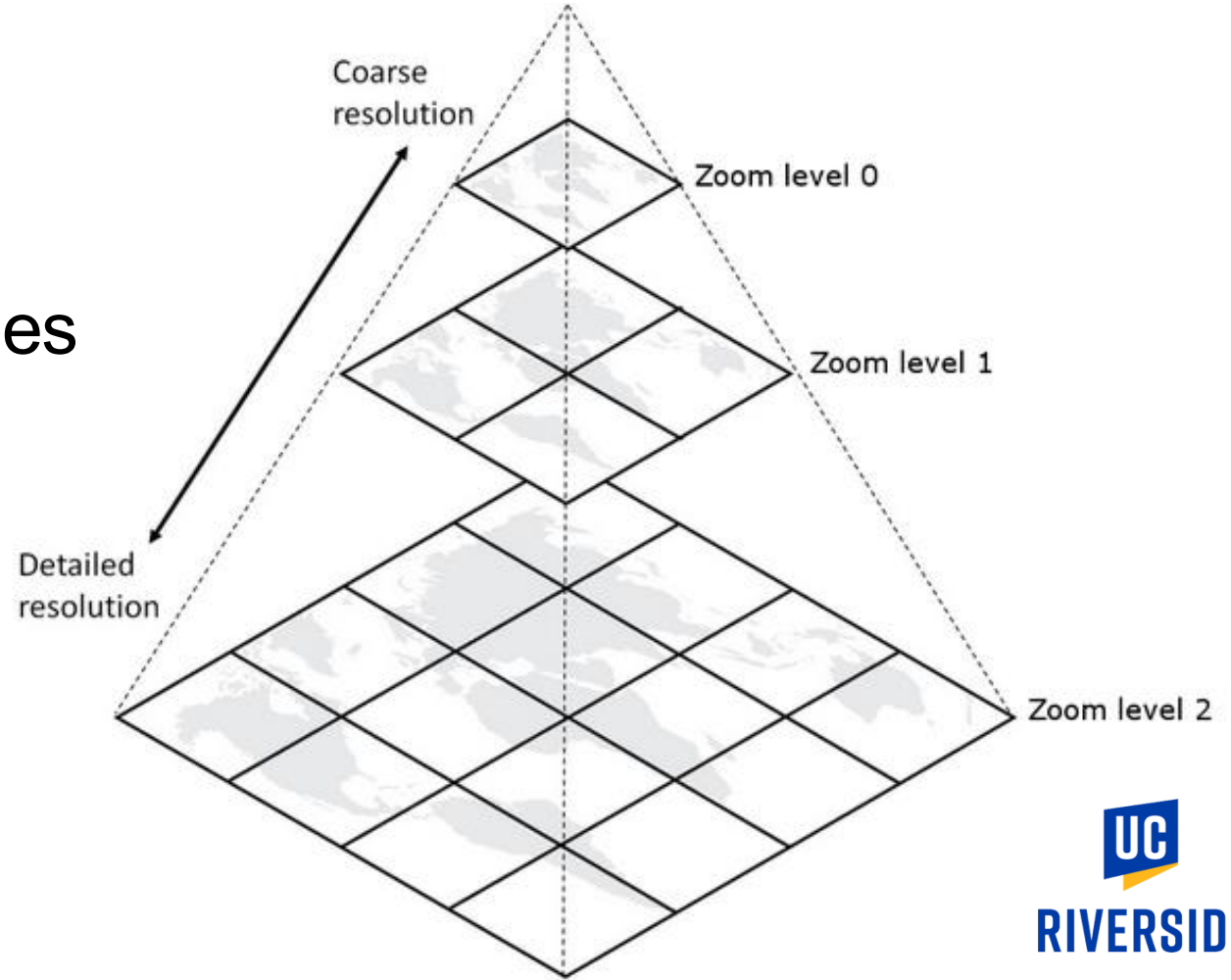
UC RIVERSIDE

Build Tile Pyramid:

- Multi-resolution grid (zoom levels)
- Each level subdivides space into tiles

Histogram is used to:

- Estimate tile sizes
- Decide which tiles to generate



UC RIVERSIDE

MVT Generation:

CLI Starlet Command:

```
starlet build --dir datasets/mydata --zoom 7 -- threshold 100000
```

For each tile:

- Clip Geometries
- Simplify Geometries
- Encode Attributes

Output: **tile-{z}-{x}-{y}.mvt**

Only generate **large tiles** so we can quickly load them in the browser.
For smaller tiles, the server will visualize them on-the-fly.

The vector tile itself is not an image; it is **compressed vector data**.

This means that creating a visually pleasing and informative map requires an additional step of **styling** and **rendering** the vector data.

UC RIVERSIDE

Full Pipeline: (Tile + MVT)

CLI Starlet Command:

```
starlet build --input data.parquet --outdir datasets/mydata
```

Example of Running this command for the countries dataset:

starlet build --input /Users/tarlan/Downloads/NE_countries.parquet --outdir datasets/NE_countries

Serve Interactive Map:

CLI Starlet Command:

```
starlet serve --dir datasets --port 8765
```

- Tiles served via **Flask server**

- Map client (MapLibre / OpenLayers):
  - Requests tiles dynamically

- Tiles that are not pre-generated will be created on-the-fly from the spatial index (histogram).

Example Python Script:

python3 serve_map.py Riverside_Parks_DominantVegetation.geojson Parks_DominantVeg

UC RIVERSIDE

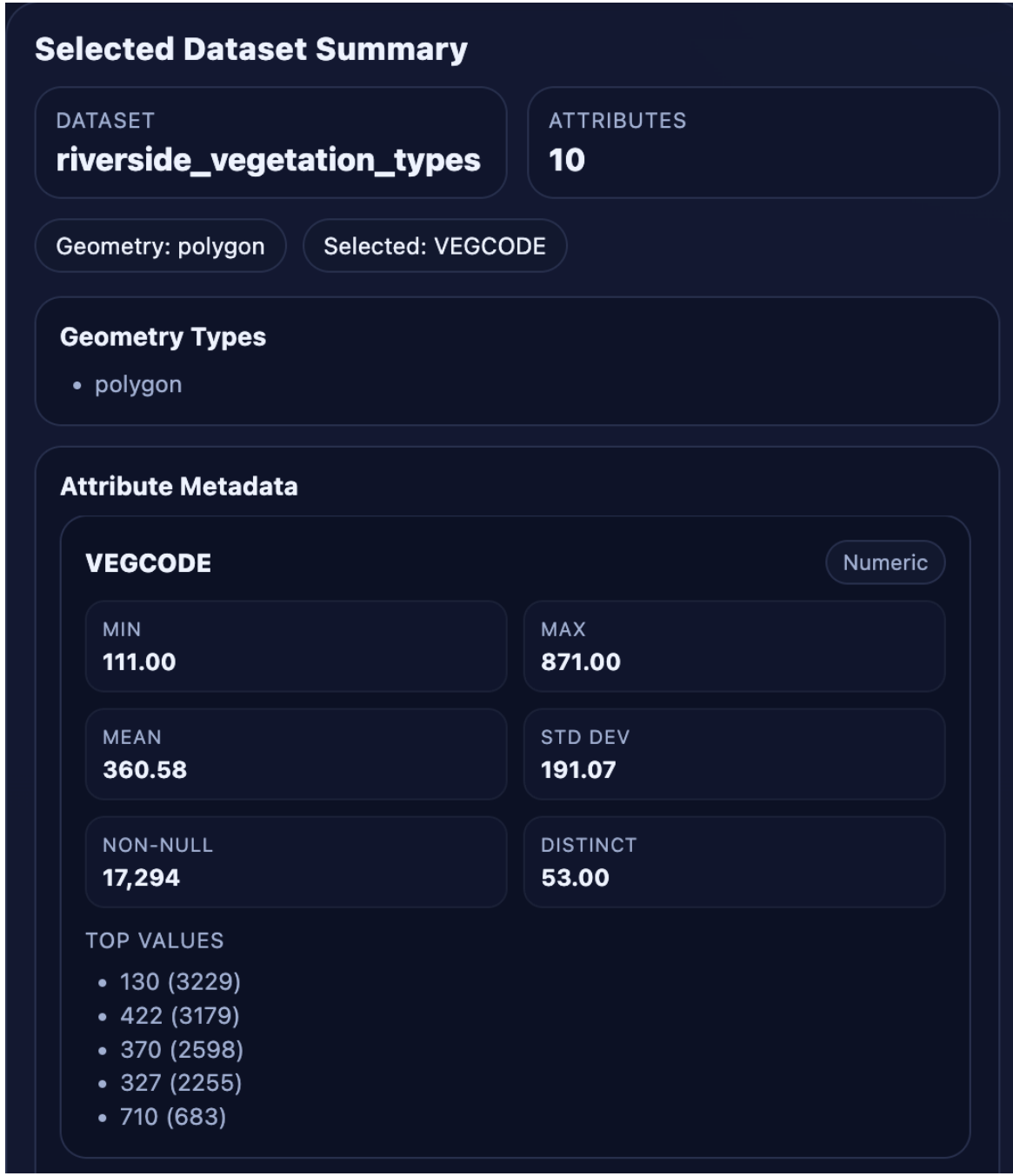
Metadata and Stats

Precomputed summaries for each attribute:

- **Numeric** → min, max, mean, stddev
- **Categorical** → top-k frequencies
- **Temporal** → time ranges

Stored alongside dataset (attributes.json)

LLM-guided dataset styling: Lightweight prompts, no need to share the complete dataset.

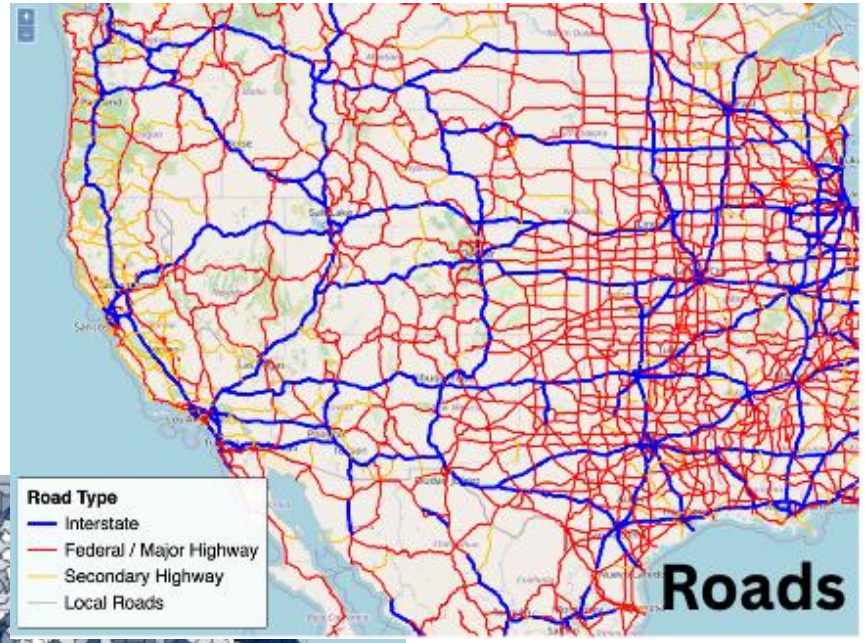


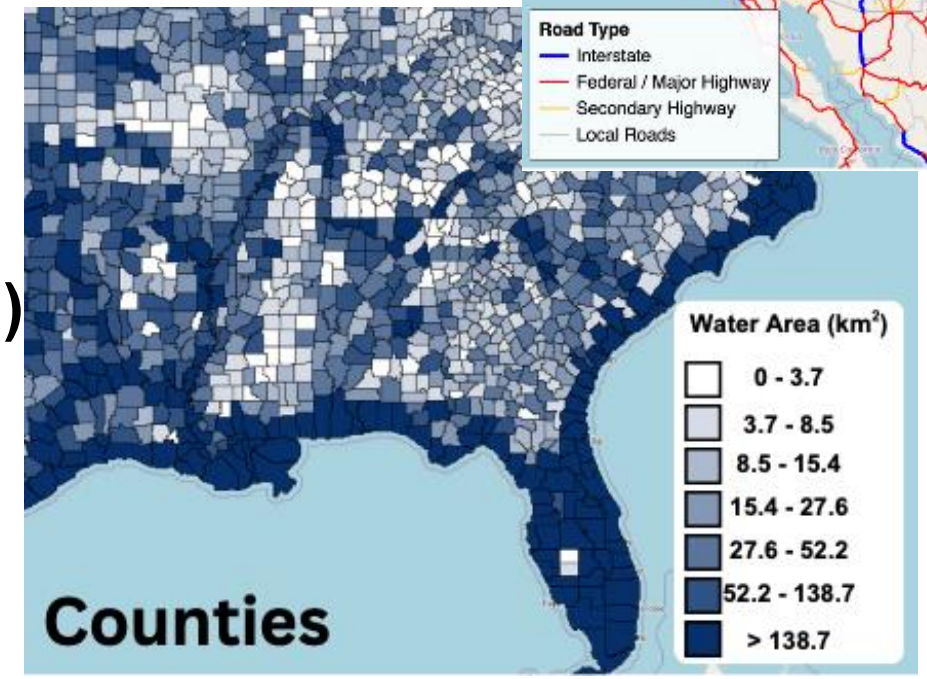
UC RIVERSIDE

Styling Rendered Map:

HTML/JavaScript Code. We will use **LLM** to help generate this code.

- Visualize data using **color, size, and opacity**
- Map attributes → visual encodings:
  - Numeric → gradients (e.g., heatmaps)
  - Categorical → distinct colors
- Styling applied **at runtime (client-side)**
- Supports **dynamic updates without regenerating tiles**



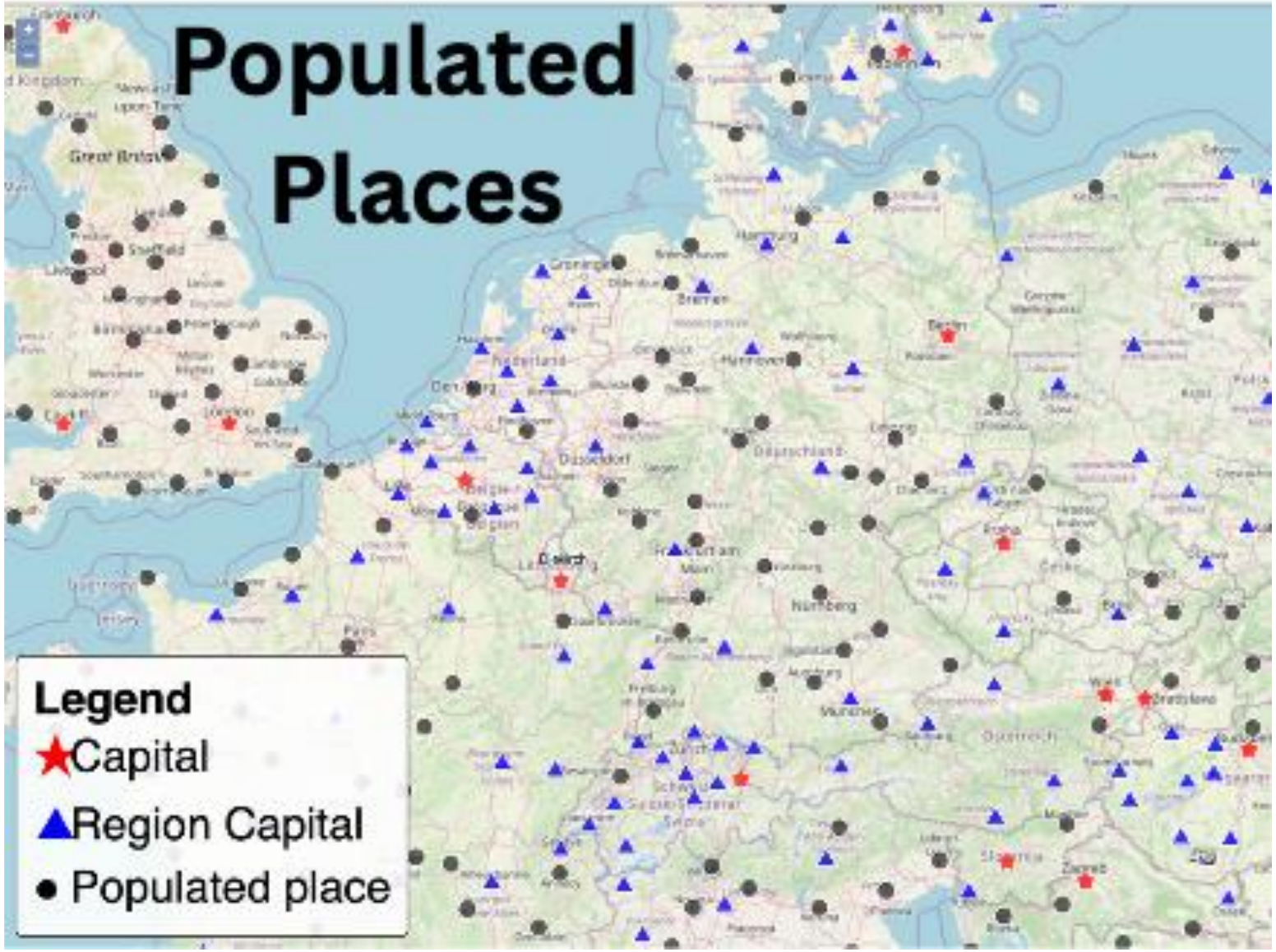


UC RIVERSIDE

More Styling Options

- **Labels:** We can display an attribute as a label, e.g., state names, or street names.

- **Icons:** Different icons represent different attribute values for type of populated place: capital, region capital or regular populated place.



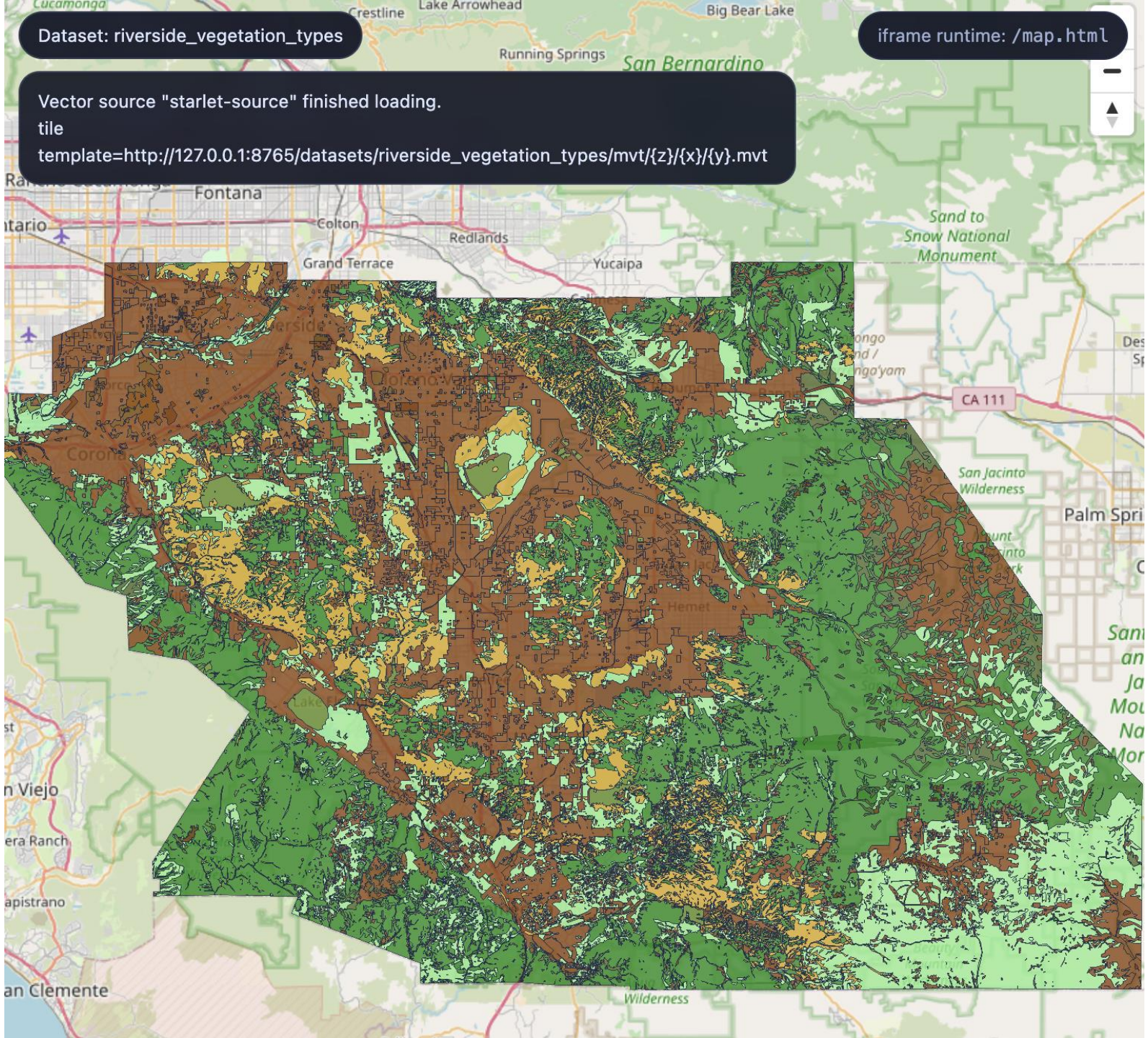
Workshop Overview

- Introduction to GIS
- Introduction to Starlet Project
- → Hands-On Example
- Introduction to LLM-Assisted Styling
- Competition

UC RIVERSIDE

Tutorial:

https://github.com/tarlaun/starmap_workshop_tutorial


Dataset: riverside_vegetation_types
iframe runtime: /map.html
Vector source "starlet-source" finished loading.
tile
template=http://127.0.0.1:8765/datasets/riverside_vegetation_types/mvt/{z}/{x}/{y}.mvt
Cucamonga
Crestline
Lake Arrowhead
Big Bear Lake
Running Springs
San Bernardino
Fontana
Colton
Redlands
Grand Terrace
Yucaipa
Sand to Snow National Monument
CA 111
San Jacinto Wilderness
Palm Spri
Hemet
Wilderness